

Shoulder — ultrasound-guided IA

NOVA-CORE 1 mL + NOVA-FLEX 1 mL · physician treatment protocol

A standardized, physician-directed framework for chronic shoulder pain, informed by a prospective 3-patient case series that reported no treatment-related adverse events and improvement in pain, mobility, stiffness, and daily function through 12 weeks after a single administration.

Indications

- Chronic shoulder pain > 6 months
- Rotator cuff tendinopathy or degenerative cuff pathology
- Mild-to-moderate glenohumeral osteoarthritis
- Chronic inflammatory shoulder pain
- Subacromial bursitis refractory to conservative treatment
- Failed PT, activity modification, medication, or prior injections

Patient preparation

1. Verify indication; review imaging when available.
2. Obtain informed consent.
3. Record baseline VAS pain score and shoulder function.
4. Position seated or lateral decubitus.
5. Prepare skin with chlorhexidine; establish sterile field.
6. Use ultrasound guidance throughout.

POST-PROCEDURE CARE

- Observe 15 minutes following injection
- Mild soreness may occur for 24–72 h
- Avoid NSAIDs and alcohol for 72 h when appropriate
- Restrict heavy overhead lifting 2–3 days; encourage gentle ROM

Contraindications

- Active local or systemic infection
- Pregnancy or breastfeeding
- Prior shoulder arthroplasty in the target joint
- Corticosteroid injection within 4 weeks
- Systemic corticosteroid use within 30 days
- Uncontrolled bleeding disorder or significant anticoagulation risk

Injection & distribution

TOTAL VOLUME

2 mL

Glenohumeral joint ≈ 1 mL; remaining volume periarticular to areas of pathology — cuff insertion, subacromial bursa, posterior capsule, or biceps tendon sheath per ultrasound findings.

Equipment: high-frequency linear transducer, sterile cover and gel, chlorhexidine, sterile gloves and drapes, 22–25 G needle.

FOLLOW-UP & OUTCOME MEASURES

Week 2

Safety review

Week 6

Functional progress

Week 12

Primary outcome

VAS PAIN

ROM

OVERHEAD TOLERANCE

ADLs

SLEEP / NIGHT PAIN

SATISFACTION

ADVERSE EVENTS

REGULATORY

Intended for physician-directed clinical and investigational use. Providers are responsible for compliance with applicable federal, state, facility, IRB, and professional practice requirements.

Today you received an ultrasound-guided injection of cell and exosome products in and around your shoulder. The products used are made from donated birth tissue and are not FDA-approved treatments. This page tells you what is normal in the days ahead, how to care for yourself, and when to call us.

DATE OF TREATMENT

SITE(S) TREATED

NEXT VISIT

WHAT TO EXPECT

Changes are gradual — do not judge the result in the first days

FIRST 72 HOURS

Aching in the shoulder and upper arm is common and usually settles in two to three days. Sleeping on that side may be uncomfortable for a few nights.

WEEKS 1–4

Keep the shoulder moving gently — pendulum swings and easy range of motion — while avoiding heavy or overhead work.

WEEKS 4–12

Progress is assessed at your follow-up visit; strength work is added back gradually as advised.

DO

- Keep the arm gently moving; a sling is not needed
- Sleep on the other side for the first few nights
- Use cold packs 10–15 minutes at a time for comfort
- Take acetaminophen (Tylenol) for soreness if needed

AVOID

- Anti-inflammatory pain relievers for 72 hours
- Alcohol for 72 hours
- Overhead lifting and reaching for 48–72 hours
- Weight training for the shoulder until cleared

CALL US RIGHT AWAY IF

For an emergency, call 911 first

- The shoulder becomes increasingly hot, red, and swollen
- New numbness, tingling, or weakness in the arm or hand
- Fever above 101 °F or shaking chills
- Severe pain not controlled by acetaminophen and cold packs

CLINIC PHONE

AFTER-HOURS PHONE

NOTES FROM YOUR VISIT